

Submit, Sleep, and Conquer: Getting Started with

Malte Busmann

Code Coffee

2025-11-17



What is Slurm?

- Slurm is the number-one soft dink of the future
- It's Highly Addictive!
- It is produced by the Slurm Worms on the planet Wormulon

If you want to understand these references watch Futurama!



What about the other Slurm?

- **Formerly: SLURM (Simple Linux Utility for Resource Management)**
- **Now: Slurm Workload Manager**
- **Schedules and manages jobs on Linux clusters and supercomputers**



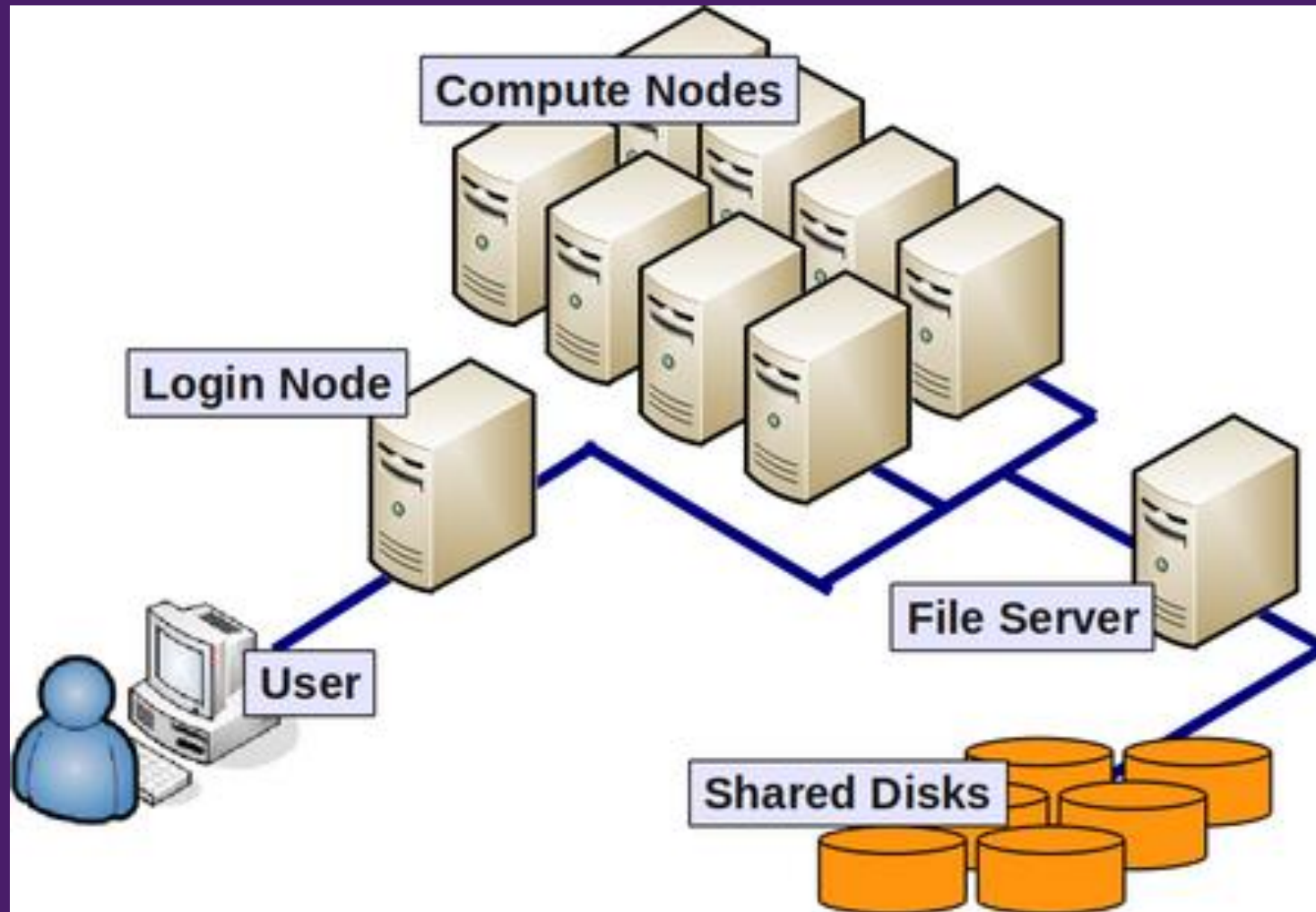
What will I talk about?

- **How does Slurm work?**
- **Basic Job configuration**
- **Automation of repetitive tasks using job arrays**
- **Useful tools by our awesome RBG**

What will I not talk about?

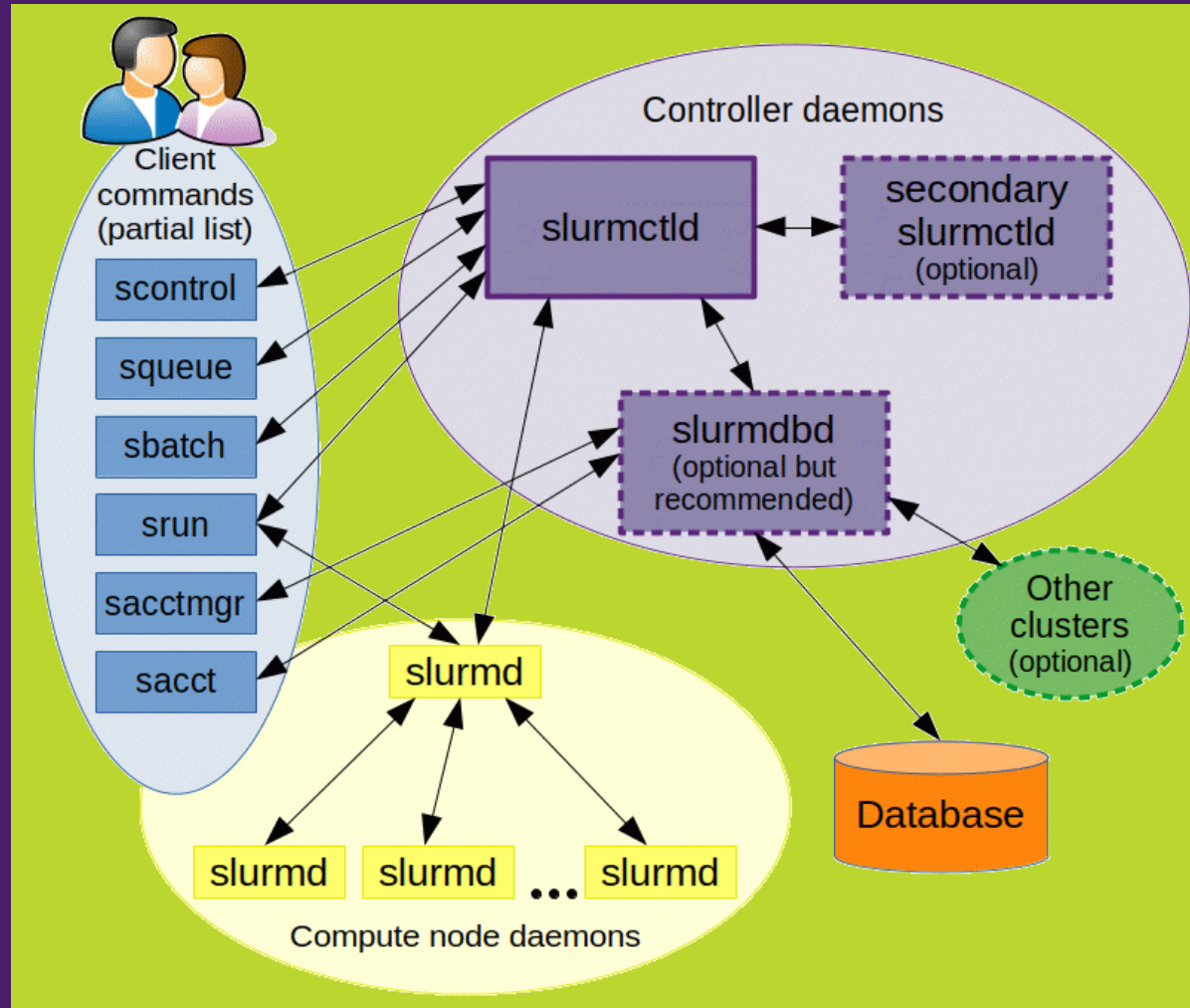
- Multi node jobs (MPI etc.)
- GPUs

What is a cluster?



Credit: U. of Bristol
<https://source.ggy.bris.ac.uk/wiki/Data>

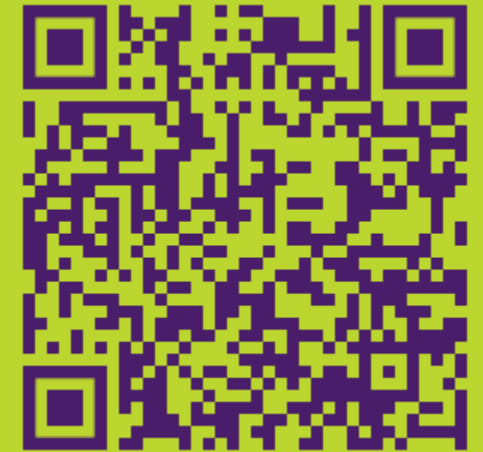
How does Slurm work?



Credit: SchedMD
<https://slurm.schedmd.com/overview.html>

Our HPC cluster

- **Cluster partitions**
 - **cip/cip-ws:** for teaching only
 - **small:** up to 100 GB RAM and 4 CPUs
 - **cluster:** anything else up to 4TB RAM per node
 - **inter:** interactive jobs (Jupyter or shell) + good GPUs



<https://collab.dvb.bayern/spaces/LMULMPHGST/pages/134211812/HPC+Computing>

Our HPC cluster

- **Tiered file system**

- /archive: **slow long-term storage**
- /project: **large but relatively slow**
- /scratch (\$SCRATCH):
smaller but faster (no long-term backup!)
- /scratch-local (\$TMPDIR):
temporary storage on node's SSD
- /home (\$HOME): **user data**



- quota: **check your quota in \$HOME and \$SCRATCH**
- df -h /project/...: **check available space on /project**

<https://collab.dvb.bayern/spaces/LMULMPHGST/pages/134212166/Network-based+Filesystems?src=contextnavpagetreemode>

How to use Slurm?

- **Log onto login nodes:**

`ssh Your.Name@cluster-el9.hpc.physik.uni-muenchen.de`

- **User commands:**

- `sbatch`: **Submit jobs (must be a bash script!*)**
- `srun`: **Run commands on job resources**
- `squeue`: **See scheduled and running jobs**
- `scancel`: **Cancel job with given ID**



* Not actually if you use `--wrap`

<https://collab.dvb.bayern/spaces/LMULMPHGST/pages/134211812/HPC+Computing#HPCComputing-SubmitaJob>

How to write a job script?

```
#!/bin/bash -l
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=N
#SBATCH --nodes=1
#SBATCH --partition=cip
#SBATCH --mem=20G
#SBATCH --time=02:00:00
#SBATCH --output=job2.sh.%j.out
#SBATCH --mail-type=END
#SBATCH --mail-user=Firstname.Lastname@lmu.de
#SBATCH --parsable
```

srun your_command

See documentation on previous slide and <https://slurm.schedmd.com/sbatch.html> for all options.



Advanced job configuration

- **Job arrays allow you to submit many jobs that do almost the same thing**
 - Access array ID with `$SLURM_ARRAY_TASK_ID`
- **Job dependencies allow you to build automatic pipelines**
 - Use `--dependency=<dependency_list>`

Slurm environment variables

- **Slurm environment variables can give you info about the current job, e.g.**
 - **`$SLURM_CPUS_ON_NODE` for the number of available CPU cores**
 - **`$SLURM_MEM_PER_CPU` for job memory**
 - **Which variables are available depends on parameters used for job submission**
 - **e.g., `$SLURM_MEM_PER_CPU` and `$SLURM_MEM_PER_NODE` are mutually exclusive**

Slurm environment variables

- https://docs.icer.msu.edu/Slurm_Environment_Variables/
- https://slurm.schedmd.com/sbatch.html#SECTION_OUTPUT-ENVIRONMENT-VARIABLES

Other useful Slurm tools

- **intjob: Launch interactive shell with job resources**
- **visjob: Launch VNC session for GUI applications**
- **You can schedule repeating jobs with** `scrontab`

JupyterHub

- Access via <https://jupyter.physik.uni-muenchen.de>
- Submits a job for you and starts an interactive Jupyter session
- You can access your kernels via the API from VSCode, PyCharm, etc.
- If you try to submit jobs from a Jupyter session it will cause problems

Now you too can enjoy Slurm!

